

# Human–AI Productivity Paradoxes

Aouad, Lykouris & Zhong (2026) — Proposition 2.1

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`github.com/johnbarraza/ai-01-aouad`

## 1. The paper

**Question.** Can AI reduce human effort while still increasing output?

**Single mechanism.** Skill, effort, and AI are perfect substitutes:

$$x = s + e + a.$$

**Agent's problem.**

$$\max_{e \geq 0} p(s + e + a) - \gamma e.$$

**Economic intuition**

Skill and AI each replace costly effort one for one. The constraint  $e \geq 0$  creates the interior-versus-corner split.

## 2. Proposition 2.1 — result and conditions

**Conditions:**  $s > 0$ ,  $a \geq 0$ ,  $\gamma > 0$ ;  $p : \mathbb{R}_+ \rightarrow \mathbb{R}_+$  is weakly increasing, weakly concave, continuous, and twice differentiable; and

$$\limsup_{x \rightarrow \infty} \frac{p(x)}{x} < \gamma.$$

Define the critical input with the **largest-maximizer tie-break**:

$$x^* = \max \arg \max_{x \geq 0} \{p(x) - \gamma x\}.$$

**Then**

$$e^*(s, a) = (x^* - s - a)_+, \quad p^*(s, a) = \max\{p(x^*), p(s + a)\}.$$

### One-sentence intuition

The worker tops total input up to  $x^*$ ; once  $s + a$  reaches that target, effort falls to zero.

### 3. What I did

1. Rewrote the choice using total input:

$$\max_{x \geq s+a} \{p(x) - \gamma x\} + \gamma(s + a).$$

2. Verified the two feasible cases instead of assuming an interior FOC:

$$x^{\text{opt}} = \begin{cases} x^*, & s + a \leq x^*, \\ s + a, & s + a > x^*. \end{cases}$$

3. Numerically audited the formula for  $p(x) = 10(1 - e^{-0.4x})$  in `verify_proposition.py`.
4. Checked possible extensions: convex effort cost is already in Appendix D; a two-period non-myopic agent remains a tractable candidate.

## 4. Where I did not believe the AI

**AI claim.** The first-order condition  $p'(s + e + a) = \gamma$  proves the result.

**Verdict:** **incomplete.**

- ▶ It starts by assuming an interior optimum.
- ▶ It misses the corner  $e = 0$ .
- ▶ It does not handle a flat maximizing set or justify the paper's largest-maximizer tie-break.

**My check.** I derived the feasible set  $x \geq s + a$  and verified both cases directly from concavity.

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